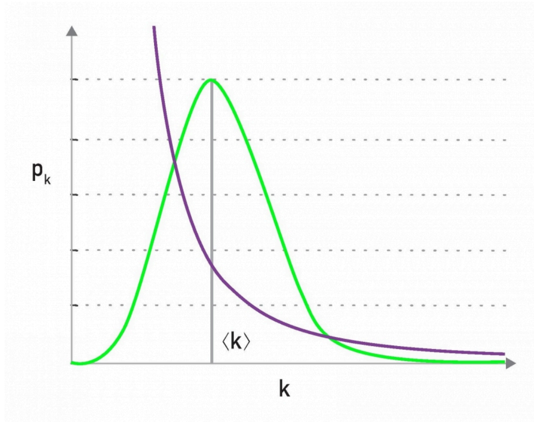
L4: Scale-free Nature of Power-law Networks



Lack of an Internal Scale, figure 4.7 from networksciencebook.com by Albert-László Barabási

One of the first things we learn in statistics is that the Normal distribution describes quite accurately many random variables *(due to the Central Limit Theorem),* and that according to that distribution 99.8% of the data are expected to fall within 3 standard deviations from the average.

Qualitatively, this is true for all distributions with exponentially fast decreasing tails, which includes the Poisson distribution and many others. In networks that have such degree distributions, the average degree represents the ***"typical scale"*** of the network, in terms of the number of connections per node *(see green distribution at the visualization).*

On the other hand, a power-law degree distribution with exponent $2 < \alpha < 3$ has finite mean but infinite variance *(any higher moments, such as skewness are also infinite).* The infinite variance of this statistical distribution means that we cannot expect the data to fall close to the mean. On the contrary, the mean *(the average node degree in our case*) can be a rather uninformative statistic in the sense that a large fraction of values can be lower than the mean, and that many values can be much higher than the mean *(see purple distribution at the visualization).*

For this reason, people often refer to power-law networks as ***"scale-free"***, in the sense that the node degree of such networks does not have a ***"typical scale".*** In the rest of this course, we prefer to use the term ***"power-law networks"*** because it is more precise.